Optimal monetary policy

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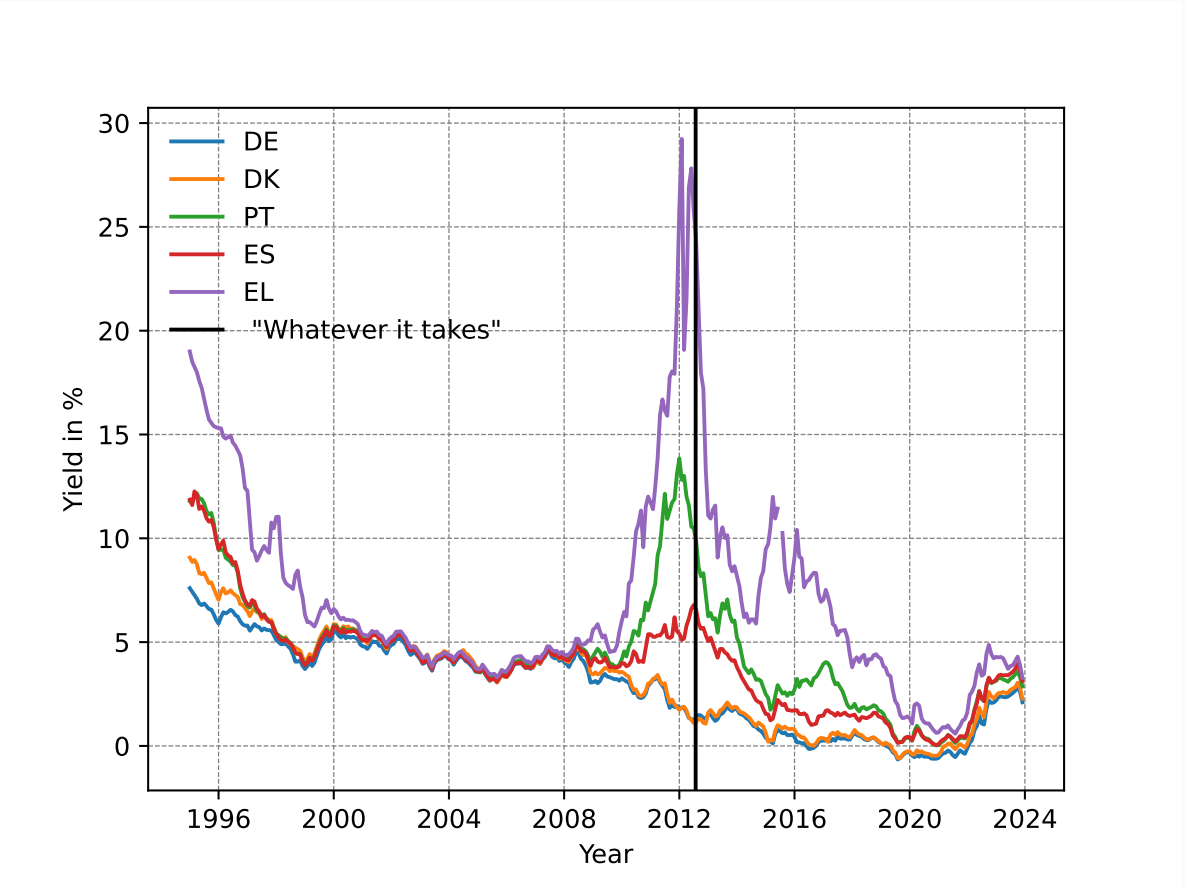
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UNIVERSITY OF COPENHAGEN

Words have consequences

Mario Draghi Video



- "Within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough."

Today

Central bankers are (rational) people, too

- Policy makers have goal functions, but where do they come from?
- Can central bankers be expected to adhere to rules?

Policy and politics

- Assuming some exogenous rule for monetary policy is too simple
- The policy itself is an outcome of its environment: it is endogenous
- In this context, credibility and reputation will play key roles

Optimal Monetary policy

- Agents and the central bank play a non-cooperative "game" ($\Longrightarrow$ Nash)
- Outcome depends on the ability of the central bank to **commit** to a plan

Monetary policy makers care about their credibility

Isabel Schnabel

- "Instead, for monetary policy to remain credible in the current environment, it must not be an inflationary source itself."

Christine Lagarde

- "What became evident [during 2012] is that the perceived commitment of policymakers was a crucial variable in effective policymaking."

Janet Yellen

- "My remarks today will focus on the issue of credibility—in particular on the Federal Reserve's credibility regarding its announced commitment to maintaining price stability."

Value judgements

Optimal monetary policy

- So far, all of our analyses have been positive
- Given the initial assumptions, they contained no value judgements, only descriptive conclusions
- Optimal monetary policy (i.e., what the central bank **should** do) requires us to do normative analysis

Rational expectations

- How trustworthy/predictable the central bank's actions are is important
- Policy outcomes differ based on commitment/discretion

The model

Setup (Persson & Tabellini, Ch 15)

- The model is as simple as possible to isolate the channels we care about: the influence of central bank policy on output and inflation
- There is a relationship between output and inflation (PC) and an IS curve (reduced form)

The goal is to

- identify a rule for monetary policy that optimizes a loss function
- analyze how the economy's aggregates change under different assumption on credibility

The insights carry over to the New Keynesian model!

Background

Unions

- In the Persson-Tabellini model, labor unions negotiate for some wage growth w such that

$$w = \omega + \pi^e$$

- Output growth, in turn, depends on the negotiated real wage:

$$x = \gamma - (w - \pi) - \varepsilon$$

- γ is a parameter, if wages are too high, output is too low
- supply shocks ε lower domestic output

Resulting output

$$x = \underbrace{(\gamma - \omega)}_{\theta} + (\pi - \pi^e) - \varepsilon$$

Equations

Demand Curve

$$\pi_t = m_t + \underbrace{v_t}_{\text{Demand shock}} + \underbrace{\mu_t}_{\text{MP shock}}$$

Phillips Curve

$$x_t = \underbrace{\theta_t}_{\text{natural rate of output}} + (\pi_t - \pi_t^e) - \underbrace{\varepsilon_t}_{\text{Supply shock}}$$

- Inflation depends on money growth, unexpected demand and monetary policy mistakes
- Output depends on unexpected inflation, supply and its natural rate
- Expected inflation is $\mathbb{E}_t[\pi_t] \equiv \pi_t^e$
- All shocks are independent and 0 in expectation

Survey

Commitment

Timing assumptions

Perfect commitment

1. Announcement of monetary rule
2. Everyone observes the natural level of output θ
3. Expectations π^e are formed, given the information about θ
4. Everyone observes v and ε
5. The central bank decides the money supply m
6. μ is realized, pinning down output x and inflation π

Consequences

- The central bank has an informational advantage (expectations are pinned down before the money supply is set)

Intuition

Monetary policy can move after expectations are formed

- This is a reduced form way to make monetary policy powerful
  $\Longrightarrow$ it can stabilize output against shocks
  $\Longrightarrow$ it can save the agents from themselves
- Monetary policy is decided every six weeks, wages are only renegotiated at longer intervals

Lucas again

- After θ realizes, only unexpected changes in monetary policy have an effect
- Moves in m can stabilize shocks to v and μ

$$\mathbb{E}_t[\pi_t|\theta] = \mathbb{E}_t[m_t|\theta]$$

$$x_t = \theta + (\underbrace{m_t + v_t + \mu_t}_{\pi_t} - \mathbb{E}_t[m_t|\theta]) + \varepsilon_t$$

Policy

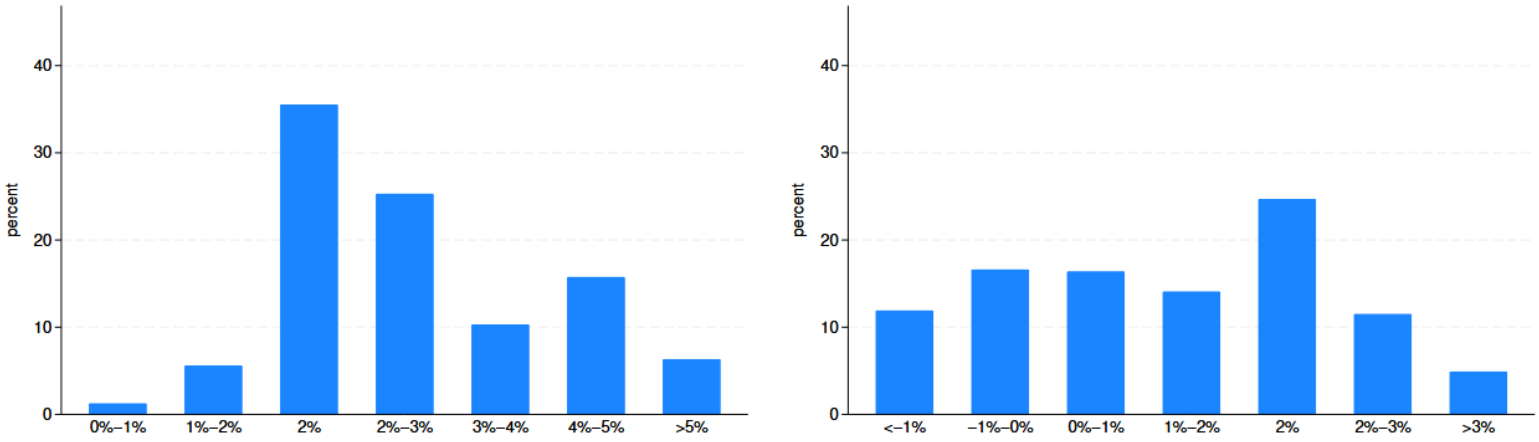
Quadratic loss function

$$\mathcal{L} = \frac{1}{2}\left[a(\pi - \overline{\pi})^2 + \lambda(x - \overline{x})^2\right]$$

- Loss function implies that the central bank dislikes deviations from some inflation benchmark $\overline{\pi}$, and deviations from some output target $\overline{x}$
- The degree of "pain" such deviations cause the banker are governed by the parameters a and λ
- The parameters a and λ are known to all agents in the economy

Subjective inflation target

Figure 3. Distribution of Perceived and Subjectively Optimal Inflation Objectives



(a) Perceived objective.

(b) Subjectively optimal inflation.

Note: "Perceived objective" refers to Question 3: "What do you think is the annual rate of inflation that the Federal Reserve is trying to achieve on average [over the five-year period starting five years from now]?" ($N = 2857$) "Optimal objective" refers to Question 7: "Now we would like you to think of the rate of inflation that would be best for the American economy [over the five-year period starting five years from now]. What do you think is this rate of inflation?" ($N = 2946$; an additional 122 respondent chose the answer "The rate of inflation does not matter for the American economy".) The 2% bin contains responses of exactly two percent. Bins above 2% exclude the lower bound of the bin range, while bins below 2% exclude the upper bound of the bin range. All distributions are weighted using Huber (1964) and sample weights.

Pfajfar–Winkler 2025

Subjective trade-offs between inflation and unemployment

Table 3: Summary Statistics for the Acceptable Sacrifice Ratio

	all values		non-negative values	
	all 2023	all 2024	2023	2024
median	.38	.25	.5	.5
mean	.38	.25	.59	.71
sd	.57	.83	.45	.6
skewness	.49	.29	.99	1.4
min	-1.2	-3.5	0	0
max	3.7	4.5	3.7	3.8
N	2070	980	1647	671

Note: The acceptable sacrifice ratio is computed as $S_i = \frac{\underline{u}_{Ai} - u_{Bi}}{\pi_{Bi} - \pi_{Ai}}$ where $u_{Bi}, \pi_{Ai}, \pi_{Bi}$ are values shown to the respondent and $\underline{u}_{Ai}$ is the answer to Question 9: "What rate of unemployment would make Scenario A equally good or bad for you and your family as Scenario B?" Statistics use Huber (1964) and sample weights.

Sacrifice ratio: the increase in the unemployment rate acceptable to reduce inflation by one percentage point. Pfajfar–Winkler 2025

Policy rule

Linear policy rule

- With a quadratic objective and linear shock processes, it can be shown that a policy rule which is linear in the shocks is optimal
- It can achieve the minimization of the loss function given the realizations of the shocks

Assumes the rule

$$m = \varphi + \varphi_\theta \theta + \varphi_v v + \varphi_\varepsilon \varepsilon$$

- The central bank reacts to shocks to natural output θ, demand shocks v and productivity shocks ε
- By definition, it cannot do anything about μ
- Recall: θ is observed before expectations are formed, v and ε realize after

Perfect credibility

Credible policy rule

- If agents know the rule and it is perfectly credible, they will include it into their expectations
- Strong assumption: central bankers may have an incentive to deviate (more on that later)

Expectations

$$\mathbb{E}_t[m_t|\theta] = \varphi + \varphi_\theta \mathbb{E}[\theta|\theta] + \varphi_v \mathbb{E}[v|\theta] + \varphi_\varepsilon \mathbb{E}[\varepsilon|\theta]$$
$$\mathbb{E}_t[\pi_t|\theta] = \mathbb{E}[m_t|\theta] = \varphi + \varphi_\theta \theta$$

- The shocks are independent $\implies$ conditional expectations don't help
- Expected inflation only depends on the realization of θ
- Other shocks are 0 in expectation

Perfect credibility

$$\mathbb{E}_t[\pi_t] = \varphi + \varphi_\theta \theta$$

Realized inflation

$$\begin{aligned}\pi_t &= \underbrace{\varphi + \varphi_\theta \theta + \varphi_v v + \varphi_\varepsilon \varepsilon}_{m_t} + v + \mu \\ &= \varphi + \varphi_\theta \theta + (1 + \varphi_v) v + \varphi_\varepsilon \varepsilon + \mu\end{aligned}$$

Realized output

$$\begin{aligned}x &= \theta + (\underbrace{\varphi + \varphi_\theta \theta + (1 + \varphi_v) v + \varphi_\varepsilon \varepsilon + \mu}_{\pi_t} - \underbrace{\varphi - \varphi_\theta \theta}_{\pi_t^e}) - \varepsilon \\ &= \theta + (1 + \varphi_v) v + (\varphi_\varepsilon - 1) \varepsilon + \mu\end{aligned}$$

What is the optimal policy?

Ex-ante optimality

- What parameters should be set for the policy rule **ex-ante**?
- CB ties its hands: **could** move after private sector but doesn't
- Crucially: Implies optimal policy **in expectation**

Minimize the loss function

- If the rule is credible, then output and inflation will behave as on the previous slide
- Plug into the loss function
- Minimize **the expectation**

Expected loss

$$
\begin{aligned}
\mathbb{E}[\mathcal{L}] &= \frac{1}{2}\mathbb{E}\left[a(\pi-\overline{\pi})^2+\lambda(x-\overline{x})^2\right] \\
&= \frac{1}{2}\mathbb{E}\Bigg[a(\underbrace{\varphi+\varphi_\theta\theta+(1+\varphi_v)v+\varphi_\varepsilon\varepsilon+\mu}_{\pi}-\overline{\pi})^2 \\
&\quad +\lambda(\underbrace{\theta+(1+\varphi_v)v+(\varphi_\varepsilon-1)\varepsilon+\mu}_{x}-\overline{x})^2\Bigg] \\
&= \frac{1}{2}\mathbb{E}\left[a(A)+\lambda(B)\right]
\end{aligned}
$$

- Want to minimize $\Longrightarrow$ take derivatives
- But: Expectation of square term is complicated, better multiply out first (next slide)

Algebra

$$
\begin{aligned}
A &= (\varphi + \varphi_\theta\theta + (1+\varphi_v)v + \varphi_\varepsilon\varepsilon + \mu - \overline{\pi})^2 \\
&= \varphi^2 + \varphi\varphi_\theta\theta + \varphi(1+\varphi_v)v + \varphi\varphi_\varepsilon\varepsilon + \varphi\mu - \varphi\overline{\pi} + \varphi\varphi_\theta\theta + \varphi_\theta^2\theta^2 + (1+\varphi_v)v\varphi_\theta\theta \\
&+ \varphi_\varepsilon\varepsilon\varphi_\theta\theta + \mu\varphi_\theta\theta - \overline{\pi}\varphi_\theta\theta + \varphi(1+\varphi_v)v + \varphi_\theta\theta(1+\varphi_v)v + (1+\varphi_v)^2v^2 \\
&+ \varphi_\varepsilon\varepsilon(1+\varphi_v)v + \mu(1+\varphi_v)v - \overline{\pi}(1+\varphi_v)v + \varphi_\varepsilon\varepsilon\varphi + \varphi_\theta\varphi_\varepsilon\varepsilon\theta + (1+\varphi_v)\varphi_\varepsilon\varepsilon v \\
&+ \varphi_\varepsilon^2\varepsilon^2 + \varphi_\varepsilon\varepsilon\mu - \varphi_\varepsilon\varepsilon\overline{\pi} + \varphi\mu + \varphi_\theta\mu\theta + (1+\varphi_v)\mu v + \varphi_\varepsilon\mu\varepsilon + \mu^2 - \mu\overline{\pi} \\
&+ \varphi\overline{\pi} + \varphi_\theta\overline{\pi}\theta + (1+\varphi_v)\overline{\pi}v + \varphi_\varepsilon\overline{\pi}\varepsilon + \overline{\pi}\mu - \overline{\pi}\overline{\pi}
\end{aligned}
$$

All shocks are independent!

- In expectation, shock terms multiplied by constants are zero, e.g. $\mathbb{E}[\varphi\varphi_\theta\theta] = \varphi\varphi_\theta\mathbb{E}[\theta] = 0$
- In expectation, cross-terms are zero: $\mathbb{E}[\varphi_\varepsilon\varepsilon\varphi_\theta\theta] = \varphi_\varepsilon\varphi_\theta\mathbb{E}[\varepsilon\theta] = \varphi_\varepsilon\varphi_\theta\mathbb{E}[\varepsilon]\mathbb{E}[\theta] = 0$

Light at the end of the tunnel

$$
\begin{aligned}
\mathbb{E}[A] &= \mathbb{E}[(\varphi + \varphi_\theta \theta + (1+\varphi_v)v + \varphi_\varepsilon \varepsilon + \mu - \overline{\pi})^2] \\
&= \varphi^2 - \varphi\overline{\pi} + \varphi_\theta^2 \mathbb{E}[\theta^2] + (1+\varphi_v)^2 \mathbb{E}[v^2] + \varphi_\varepsilon^2 \mathbb{E}[\varepsilon^2] + \mathbb{E}[\mu^2] - \varphi\overline{\pi} + \overline{\pi}^2 \\
&= \varphi^2 - \varphi\overline{\pi} + \varphi_\theta^2 \sigma_\theta^2 + (1+\varphi_v)^2 \sigma_v^2 + \varphi_\varepsilon^2 \sigma_\varepsilon^2 + \sigma_\mu^2 - \varphi\overline{\pi} + \overline{\pi}^2
\end{aligned}
$$

Expectations depend on variances of shocks

- $\sigma_q^2 = \mathrm{Var}(q) = \mathbb{E}[(q - \overline{q})^2]$ – If mean of random variable is 0, the expectation of its square is the variance
- The zero-mean and independence assumptions are doing **a lot** of heavy lifting for us

Second square term

Apply the same principle to the square variable B

$$\begin{aligned}\mathbb{E}[B] &= \mathbb{E}[(\theta + (1+\varphi_v)v + (\varphi_\varepsilon - 1)\varepsilon + \mu - \overline{x})^2] \\ &= \mathbb{E}[\theta^2] + (\varphi_v + 1)^2\mathbb{E}[v^2] + (1-\varphi_\varepsilon)^2\mathbb{E}[\varepsilon^2] + \mathbb{E}[\mu^2] + \overline{x}^2 \\ &= \sigma_\theta^2 + (\varphi_v + 1)^2\sigma_v^2 + (1-\varphi_\varepsilon)^2\sigma_\varepsilon^2 + \sigma_\mu^2 + \overline{x}^2\end{aligned}$$

- Now we have all the ingredients to fill in the expectation of the loss function

Minimize expected loss

Plugging in from the previous slides:

$$\min_{\varphi,\varphi_\theta,\varphi_v,\varphi_\varepsilon} \mathbb{E}[\mathcal{L}] = \frac{1}{2} a \left(\varphi^2 - \varphi\overline{\pi} + \varphi_\theta^2 \sigma_\theta^2 + (1+\varphi_v)^2 \sigma_v^2 + \varphi_\varepsilon^2 \sigma_\varepsilon^2 + \sigma_\mu^2 - \varphi\overline{\pi} + \overline{\pi}^2 \right)$$
$$+ \frac{1}{2} \lambda \left(\sigma_\theta^2 + (\varphi_v + 1)^2 \sigma_v^2 + (1-\varphi_\varepsilon)^2 \sigma_\varepsilon^2 + \sigma_\mu^2 + \overline{x}^2 \right)$$

- A hypothetical social planner wants to set the rule (i.e., the parameters in the central bank's response function) to minimize this loss
- The rule is in place forever $\implies$ minimizing single period expectation of loss is the same as discounted infinite sum of all future periods' losses
- Take the derivatives w.r.t. $\varphi, \varphi_\theta, \varphi_v, \varphi_\varepsilon$

Minimum expected loss

$$\frac{\partial \mathbb{E}[\mathcal{L}]}{\partial \varphi} : a(\varphi - \overline{\pi}) = 0 \implies \varphi = \overline{\pi}$$
$$\frac{\partial \mathbb{E}[\mathcal{L}]}{\partial \varphi_\theta} : a\vartheta_\theta \sigma_\theta^2 = 0 \implies \varphi_\theta = 0$$
$$\frac{\partial \mathbb{E}[\mathcal{L}]}{\partial \varphi_v} : \sigma_v^2(a + \lambda)(1 + \varphi_v) = 0 \implies \varphi_v = -1$$
$$\frac{\partial \mathbb{E}[\mathcal{L}]}{\partial \varphi_\varepsilon} : \sigma_\varepsilon^2(a\varphi_\varepsilon - \lambda(1 - \varphi_\varepsilon)) = 0 \implies \varphi_\varepsilon = \frac{\lambda}{a + \lambda}$$

Implications

- Anchor inflation where society wants it
- Shocks that are priced into expectations need no reaction (θ)
- Neutralize demand shocks
- Supply shocks: it depends. Countering supply shocks causes less deviations from $\overline{x}$, but at the cost of deviations from $\overline{\pi} \implies$ tradeoff

Equilibrium under commitment

Optimal rule

$$m_t = \overline{\pi} - v_t + \frac{\lambda}{a+\lambda}\varepsilon$$

Equilibrium inflation

$$\pi^C = \overline{\pi} + \frac{\lambda}{a+\lambda}\varepsilon + \mu$$

Equilibrium output

$$x^C = \theta - \frac{a}{a+\lambda}\varepsilon + \mu$$

- Output may fluctuate due to changes in the natural rate, supply shocks or policy errors
- Inflation only changes due to supply shocks and policy errors
- Depending on preferences, supply shocks will feed more into output, or more into inflation

Commitment conclusions

Benefits

- If the policy maker can commit to a rule, inflation and output are stable around their natural levels
- As we will see, this is the best possible outcome

Simplify the problem

- Demand shocks are neutralized $\Longrightarrow$ we can ignore them
- Policy errors μ are not interesting to study because there is little we can do about them $\Longrightarrow$ ignore for now
- **The only important shocks left are θ and ε**

Credibility

Problems with rules

- Central bankers are not computers. They may want to exploit their informational advantage
- Once expectations are locked in, it's possible to decrease societal losses even further
- The rule may not be credible if bankers have **discretion** (i.e., ability) to deviate

Discretion

- It's more realistic to assume policy makers don't stick to a rule
- This feeds back into agents (rational) expectations
- Equilibrium outcomes are different without commitment

Not required to leave!

Credibility

Opinion **Instant Insight**

Jay Powell is defiant in the face of Trump's threats

The Fed chair is determined to see out his term, and lawyers at the central bank think he has good grounds to do so



GILLIAN TETT + Add to myFT



Even if Jay Powell's confidence is breached by the Supreme Court, he knows that investors — and most US politicians — back central bank independence © Bloomberg

Gillian Tett

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Discretion

New timing

Discretion/Non-credible rule

1. ~~Announcement of monetary rule~~
2. Everyone observes the natural level of output θ
3. Expectations π^e are formed, given the information about θ
4. Everyone observes ε
5. The central bank decides the money supply m
6. Output x and inflation π are pinned down

Implications

- Without a (credible) rule, the central bank is free to do what it wants each period

New optimality

Ex-post optimality

- When the CB could commit to a credible rule, that rule was **ex-ante** optimal: $\frac{\partial \mathbb{E}[\mathcal{L}]}{\partial m} = 0$
- Without a rule, policy will be ex-post optimal: $\frac{\partial \mathcal{L}}{\partial m} = 0$
- What seems like a small difference has big consequences

Nash-equilibrium

- Central bank and consumers play a game. In equilibrium nobody wants to deviate from decision
- Solve by backwards induction

Central bank optimum under discretion (second stage)

$$\mathcal{L} = \frac{1}{2}\left[a(\pi_t - \overline{\pi})^2 + \lambda(x_t - \overline{x})^2\right]$$

$\pi_t = m_t$ remember: v and μ set to 0

$$x_t = \theta + (\pi_t - \pi_t^e) - \varepsilon_t$$

- Since $\pi_t = m_t$, just assume that the CB sets π_t directly
- Recall that the CB takes π_t^e as given

$$\frac{\partial \mathcal{L}}{\partial \pi_t}: \qquad a(\pi_t - \overline{\pi}) + \lambda(\theta + (\pi_t - \pi_t^e) - \varepsilon_t - \overline{x}) = 0$$

$$\Longrightarrow \pi = \frac{a}{a+\lambda}\overline{\pi} + \frac{\lambda}{a+\lambda}(\pi_t^e - \theta + \varepsilon + \overline{x})$$

- Note: If we plug in the result from the commitment equilibrium $\pi = \pi^e = \overline{\pi}$, $\frac{\partial \mathcal{L}}{\partial \pi_t} > 0 \Longrightarrow$ CB can do better!

Consumer expectation under discretion (first stage)

Take expectation of central banks decision function

$$\begin{aligned}\mathbb{E}[\pi|\theta] &= \frac{a}{a+\lambda}\overline{\pi} + \frac{\lambda}{a+\lambda}\mathbb{E}[(\mathbb{E}[\pi|\theta] - \theta + \varepsilon + \overline{x})|\theta] \\ &= \overline{\pi} + \underbrace{\frac{\lambda}{a}(\overline{x} - \theta)}_{\text{Inflation bias}}\end{aligned}$$

- Expected inflation is higher than in the commitment case
- Because θ is known to consumers, they know what the CB will do. If $\theta < \overline{x}$: increase m, if $\theta > \overline{x}$: decrease m
- However, these actions are pointless, because prices adjust. As always, if higher m is expected, p (and therefore π) adjusts, and x stays constant

Realized values of inflation and output

Output

$$x^D = \theta - \frac{a}{a+\lambda}\varepsilon$$

- Output is the same as under commitment!

Inflation

$$\pi^D = \overline{\pi} + \frac{\lambda}{a}(\overline{x} - \theta) + \frac{\lambda}{a+\lambda}\varepsilon$$

- Inflation is higher and more volatile

Giving central banks discretion leaves output constant, but $\mathcal{L}$ is actually higher than it could be under commitment

Realized values of inflation and output

US Treasury bonds + Add to myFT

Trump's war with Fed threatens lasting damage to Treasuries, say investors

Big investors warn of risks to world's largest and most liquid bond market from US president's attacks on central bank



Growing worries about central bank independence has fuelled a sell-off in Treasuries © Kent Nishimura/Bloomberg

Ian Smith in London and **Kate Duguid** in New York

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Institutions

Realized values of inflation and output

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FirstFT: Markets rise after Trump says he has 'no intention' of firing Powell

Also in today's newsletter, Putin's Ukraine proposal and Maga Catholics seek 'Trump-like' pope



Markets rallied after Donald Trump softened some of his criticism of Fed chair Jay Powell © AFP via Getty Images

Gordon Smith, **Tee Zhuo** and **Benjamin Wilhelm**

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Policy mandates

European Central Bank

- The primary objective of the European System of Central Banks (hereinafter referred to as 'the ESCB') shall be to maintain price stability. (Article 127, TFEU)
- In pursuing price stability, the ECB seeks to hold inflation below but close to 2 percent over a medium-term horizon.
- (...) support the general economic policies in the Union with a view to contributing to the achievement of the objectives of the Union as laid down in Article 3 of the Treaty on European Union.

Federal Reserve

- (...) so as to promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates.

Central bank appointments

Guidance

- Both ECB and Fed have been given clear guidelines on what to focus their policies on
- ECB: Price stability – everything else is secondary
- Fed: Dual mandate – more in line with the formulas above

Doves or Hawks

- However, it is impossible for a central bank to credibly commit to a rule
- Still, governments can at least appoint the right person to head the central bank
- Who are they?

Finding the right central banker

Central banker loss function

$$\mathcal{L}_{\mathcal{B}} = \frac{1}{2}\left[a_B(\pi_t - \overline{\pi})^2 + \lambda_B(x_t - \overline{x})^2\right]$$

Discretionary solution

$$x_B^D = \theta - \frac{a_B}{a_B + \lambda_B}\varepsilon$$

$$\pi_B^D = \overline{\pi} + \frac{\lambda_B}{a_B}(\overline{x} - \theta) + \frac{\lambda_B}{a_B + \lambda_B}\varepsilon$$

- Each central banker has their own a_B and λ_B
- Which one should be chosen to make decisions?

$\Longrightarrow$ analyze conditional loss function

Society's loss

Society's loss function **given** x and π with central banker

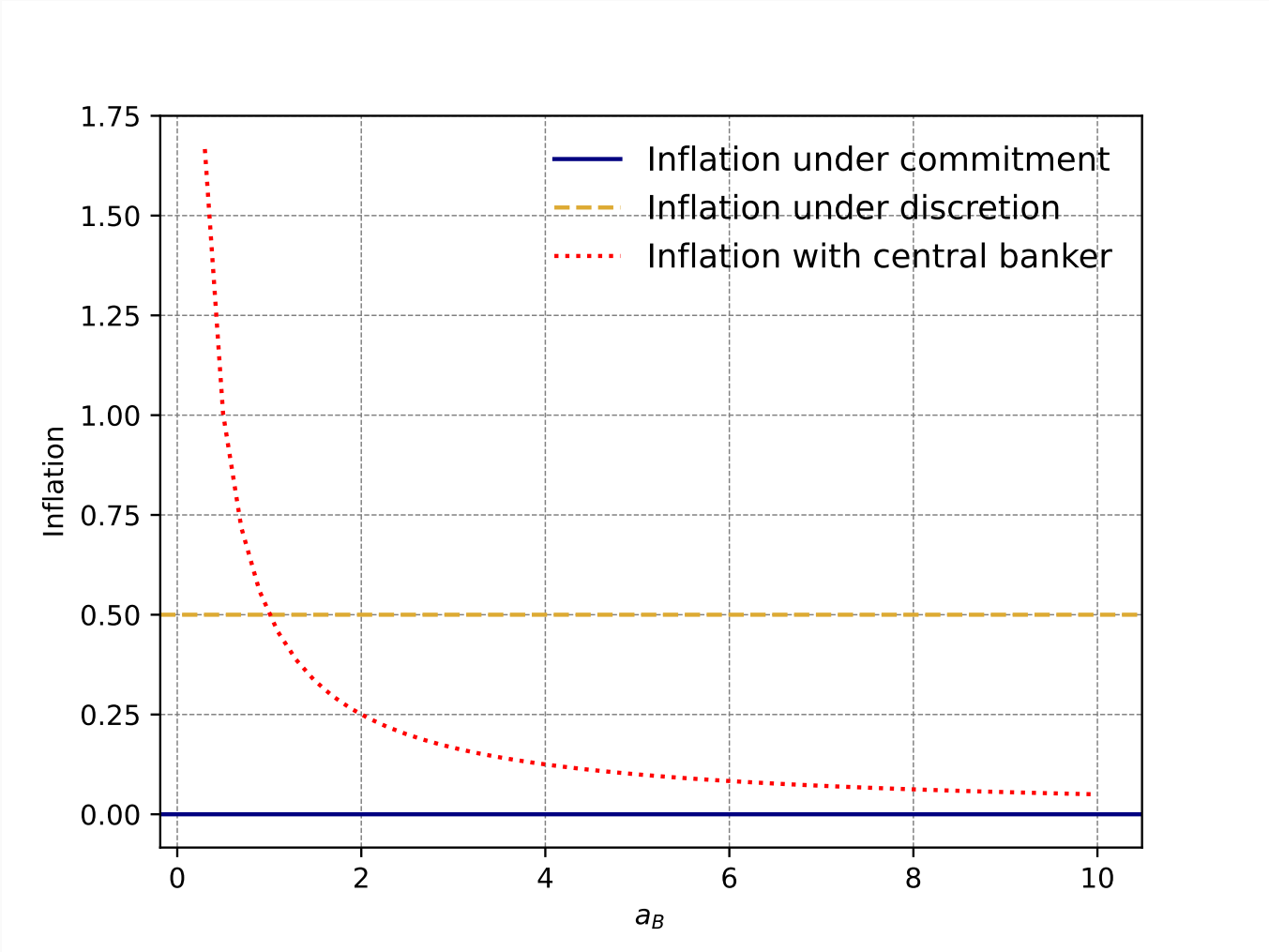
$$\mathbb{E}\left[\mathcal{L}(x_B^D, \pi_B^D)\right] = \mathbb{E}\left[\frac{1}{2}a\left(\frac{\lambda_B}{a_B}(\overline{x}-\theta)+\frac{\lambda_B}{a_B+\lambda_B}\varepsilon\right)^2+\frac{1}{2}\lambda\left(\theta-\frac{a_B}{a_B+\lambda_B}\varepsilon-\overline{x}\right)^2\right]$$

$$= \frac{1}{2}\left[a\left(\frac{\lambda_B^2}{a_B^2}(\overline{x}^2+\sigma_\theta^2)+\frac{\lambda_B^2}{(a_B+\lambda_B)^2}\sigma_\varepsilon^2\right)+\lambda\left(\sigma_\theta^2+\frac{a_B^2}{(a_B+\lambda_B)^2}\sigma_\varepsilon^2\right.\right.$$

$$= \frac{1}{2}\left[\left(\frac{\lambda_B^2+a_B^2}{a_B^2}(\overline{x}^2+\sigma_\theta^2)+\frac{\lambda_B^2+a_B^2}{(a_B+\lambda_B)^2}\sigma_\varepsilon^2\right)\right]$$

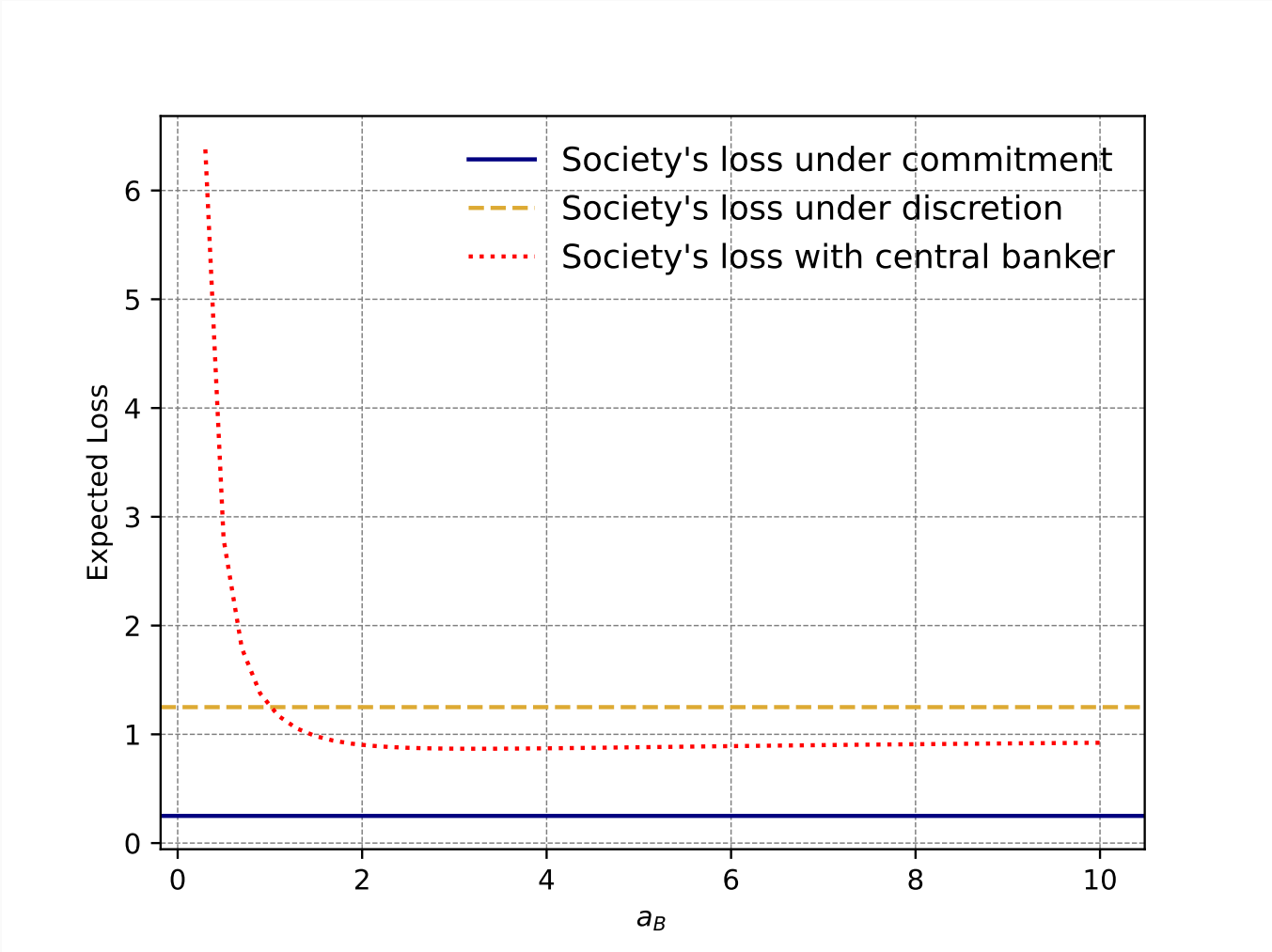
- Set society's preferences $\lambda = a = 1$ just for simplicity

Finding the right central banker



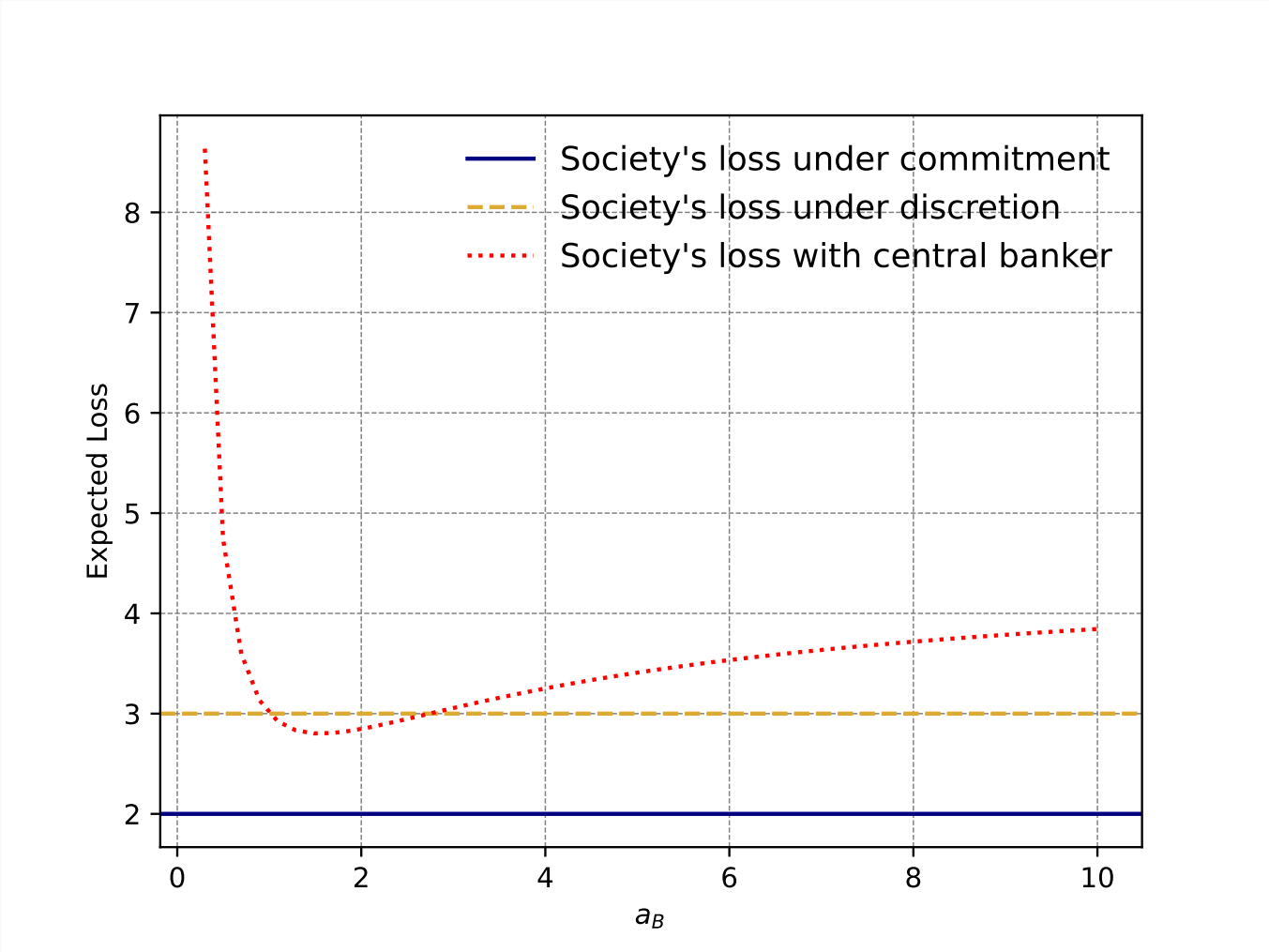
- Inflation hawks keep inflation low after hypothetical θ

Finding the right central banker



- Inflation hawks minimize the **expected** loss function

Finding the right central banker with large supply shocks



- Inflation hawks minimize the **expected** loss function
- Large supply shocks create a trade-off

Conclusion

Commitment and Discretion

- Commitment to a rule leads to lowest inflation
- Discretion creates an inflationary bias—output is unchanged

$\Longrightarrow$ Credibility is important

Repeated game

- Interaction across many periods can keep the central bank in check
- Future costs of deviating make optimum more attractive

The ideal central banker

- Inflation hawks lead to a lower loss function
- Can approach commitment optimum if there are no supply shocks

Appendix

Algebra

Back

$$\sum_{t=s+1}^{s+T} \beta^{t-s} \left(\frac{1}{2}\lambda^2 - \lambda \mathbb{E}[(\theta - \varepsilon)] \right) + \sum_{t=s+1}^{s+T} \beta^{t-s} \lambda \mathbb{E}\left[(\theta - \varepsilon)\right]$$

$$= \sum_{t=s+1}^{s+T} \beta^{t-s} \frac{1}{2}\lambda^2 = \frac{1}{2}\lambda^2 \sum_{t=s+1}^{s+T} \beta^{t-s}$$

$$= \frac{1}{2}\lambda^2 \sum_{t=s+1}^{s+T} \beta^{t-s} = \frac{1}{2}\lambda^2 \beta \sum_{t=s}^{s+T-1} \beta^{t-s} = \frac{1}{2}\lambda^2 \beta \sum_{j=0}^{T-1} \beta^{j}$$

$$= \frac{1}{2}\lambda^2 \beta \left(\sum_{j=0}^{\infty} \beta^{j} - \sum_{j=T-1}^{\infty} \beta^{j} \right) = \frac{1}{2}\lambda^2 \beta \left(\frac{1}{1-\beta} - \sum_{j=T-1}^{\infty} \beta^{j} \right)$$

$$= \frac{1}{2}\lambda^2 \beta \left(\frac{1}{1-\beta} - \beta^{T-1} \sum_{j=0}^{\infty} \beta^{j} \right)$$

$$= \frac{1}{2}\lambda^2 \beta \frac{1-\beta^{T-1}}{1-\beta}$$

Reputation

Longer time horizon

Single period

- The commitment and discretion cases before are single-period games
- In reality, central banks make decisions all the time

$\Longrightarrow$ Current decisions affect future reputation

Multi-period game

- The central bank makes decisions every period, proclaiming a rule
- Consumers decide whether they trust the bank or not
- Trust can never be rebuilt (strong assumption)

Longer run optimality

Infinite loss function

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^j \mathcal{L}(\pi_{t+j}, x_{t+j})$$

- The central bank now cares about all future periods

Simplifying assumption

$$\mathcal{L}(\pi_t, x_t) = \frac{\pi_t^2}{2} - \lambda x$$

$$\implies \pi^C = 0, \quad \pi^D = \lambda, \quad x^C = x^D = \theta - \varepsilon$$

- As always: this contains the most important intuition
- Using the double square loss function is much more messy
- Inflation volatility is costly $\implies$ CB let's ε only affect output

Betraying trust

Inflation expectations

$$\pi_t^e = 0 \text{ if } \pi_{t-1} = \pi_{t-1}^e$$

$$\pi_t^e = \lambda \text{ otherwise}$$

- If realized inflation was in line with the agents expectations yesterday, the bank has not deviated from its rule
- In this case: keep trusting the central bank
- In any other case the bank has deviated $\Longrightarrow$ don't trust the CB ever again

The central bank's problem

Adhere to rule or break trust?

- Each period, the CB faces a choice
- If it deviates, it can decrease its loss function today
- But at the cost of never being able to do so ever again

Determinants of decision

- Because the CB is a rational agent, it computes the one-time benefits of deviating and compares the to the future costs
- Whichever is more attractive is the equilibrium outcome

Contemporary benefit of deviating

Loss in case of exploitation

$$\pi_t = \lambda \qquad \text{(inflation rises)}$$

$$x_t = \theta + \lambda - \varepsilon \qquad \text{(output rises)}$$

$$\mathcal{L}(\lambda, \theta + \lambda - \varepsilon) = +\frac{1}{2}\lambda^2 - \lambda(\theta + \lambda - \varepsilon)$$

Loss in case of continuous commitment

$$\pi_t = 0$$

$$x_t = \theta - \varepsilon$$

$$\mathcal{L}(0, \theta - \varepsilon) = -\lambda(\theta - \varepsilon)$$

One-time loss from deviating

$$\mathcal{L}(\lambda, \theta + \lambda - \varepsilon) - \mathcal{L}(0, \theta - \varepsilon) = -\frac{1}{2}\lambda^2 \text{ (loss is lower)}$$

Long-run cost of deviating

Loss in case of deviating (starting at period $t = s + 1$—tomorrow)

$$\pi_s = \lambda, \quad x_s = \theta - \varepsilon$$

$$\mathbb{E}\left[\sum_{t=s+1}^{s+T} \beta^{t-s} \mathcal{L}(\lambda, \theta - \varepsilon)\right] = \sum_{t=s+1}^{s+T} \beta^{t-s} \left(\frac{1}{2}\lambda^2 - \lambda \mathbb{E}[(\theta - \varepsilon)]\right)$$

Loss in case of continuous commitment

$$\pi_s = 0, \quad x_s = \theta - \varepsilon$$

$$\mathbb{E}\left[\sum_{t=s+1}^{s+T} \beta^{t-s} \mathcal{L}(0, \theta - \varepsilon)\right] = -\sum_{t=s+1}^{s+T} \beta^{t-s} \lambda \mathbb{E}\left[(\theta - \varepsilon)\right]$$

Long-run loss from deviating Algebra

$$\mathbb{E}\left[\sum_{t=s+1}^{s+T} \beta^{t-s} \mathcal{L}(\lambda, \theta - \varepsilon)\right] - \mathbb{E}\left[\sum_{t=s+1}^{s+T} \beta^{t-s} \mathcal{L}(0, \theta - \varepsilon)\right] = \beta \frac{1}{2} \frac{(1 - \beta^{T-1})}{1 - \beta} \lambda^2$$

Overall cost-benefit analysis

Add up single-period and long-run losses from deviating

$$Q = \beta \frac{1}{2} \frac{(1 - \beta^{T-1})}{1 - \beta} \lambda^2 - \frac{1}{2} \lambda^2$$

- If $Q < 0$, the effect of deviating on the loss function (contemporaneous + long-run) is negative $\implies$ desirable! Smaller loss means gain
- If $Q > 0$, the loss is positive (that's bad) and the CB does not want to deviate
- The central bank will deviate if:

$$\frac{1}{2} \lambda^2 \left(\beta \frac{(1 - \beta^{T-1})}{1 - \beta} - 1 \right) < 0 \iff \beta \frac{(1 - \beta^{T-1})}{1 - \beta} < 1$$

Intuition

$$\beta \frac{(1-\beta^{T-1})}{1-\beta} < 1$$

Special cases

- If the world end tomorrow ($T = 1$), $0 < 1$ implies that the central bank will deviate with certainty
- If the world never ends, we need $\beta < 0.5$ for the CB to find deviating attractive
- If the discount factor is low (0.5 is very low), the CB doesn't care about the future and will deviate

Implications

- The repeated game nature of this example, together with the threat of higher inflation forever, keep the central bank honest
- Once the CB has deviated, the economy can never go back